Higher Order ODEs

1 Linear ODEs

1.1 General Theory

Theorem 1.1 (Existence and Uniqueness Theorem).
Let $I \subseteq \mathbb{R}$ be an open interval and suppose functions $g, p_0, p_1, \ldots, p_{n-1}$ are continuous on I. Then for $t_0 \in I$ and real numbers $y_0, y_1, \ldots, y_{n-1}$, there exists exactly one solution to the IVP

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y = g(t),$$
$$y(t_0) = y_0, \quad y'(t_0) = y_1, \quad \ldots, \quad y^{(n-1)}(t_0) = y_{n-1}.$$

Theorem 1.2 (Principle of Superposition).
If $y_1, y_2, \ldots, y_n$ are solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y = 0,$$

then any linear combination

$$y = c_1y_1 + c_2y_2 + \cdots + c_ny_n,$$

where $c_1, c_2, \ldots, c_n$ are constants, is also a solution to the homogeneous equation.

Definition 1.1 (Wronskian).
The Wronskian of n functions $y_1, y_2, \ldots, y_n$ is defined as the determinant

$$W(y_1, y_2, \ldots, y_n)(t) = \begin{vmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ y_1'(t) & y_2'(t) & \cdots & y_n'(t) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{vmatrix}.$$

Definition 1.2 (Fundamental Set of Solutions).
The set of n solutions $y_1, y_2, \ldots, y_n$ to an n-th order homogeneous linear ODE defined on an interval I is called a **fundamental set of solutions (FSS)** if every solution to the homogeneous equation can be expressed as a linear combination of $y_1, y_2, \ldots, y_n$.

Theorem 1.3.
If $p_0, \ldots, p_{n-1}$ are continuous on an interval I and $y_1, \ldots, y_n$ are solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y = 0,$$

then every solution to the ODE can be expressed as a linear combination of $y_1, \ldots, y_n$ if and only if

$$\exists t \in I \text{ s.t. } W(y_1, \ldots, y_n)(t) \neq 0$$

Theorem 1.4.
Let $y_1, \ldots, y_n$ be the solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y = 0, \quad \text{for} \quad t \in I.$$

Then

$$y_1, \ldots, y_n \text{ are linearly independent} \quad \iff \quad \exists t \in I \text{ s.t. } W(y_1, \ldots, y_n)(t) \neq 0.$$

Theorem 1.5 (Abel's Theorem)**.**
If $y_1, \dots, y_n$ are solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t) y^{(n-1)} + p_{n-2}(t) y^{(n-2)} + \dots + p_1(t) y' + p_0(t) y = 0, \quad \text{for} \quad t \in I,$$

then the Wronskian is given by

$$W(y_1, \dots, y_n)(t) = c \cdot \exp\left(- \int p_{n-1}(t) \, \mathrm{d}t \right) \qquad \text{where } c \text{ is a constant depending on } y_1, \dots, y_n.$$

1.2 Homogeneous Equations with Constant Coefficients

For constants $a_n \neq 0, a_{n-1}, \dots, a_1, a_0 \in \mathbb{R}$,

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$$

Characteristic polynomial:

$$\begin{aligned} Z(r) &= a_n r^n + a_{n-1} r^{n-1} + \dots + a_1 r + a_0 \\ &= a_n (r - r_1)(r - r_2) \cdots (r - r_n) \\ &= a_n (r - r_1)^{m_1} (r - r_2)^{m_2} \cdots (r - r_k)^{m_k} \end{aligned}$$

where $r_1, r_2, \dots, r_n \in \mathbb{C}$ are the roots of $Z(r)$.

Complex-valued solutions

Complex roots must appear in conjugate pairs, i.e., if $r = \lambda + \mathrm{i}\mu$ is a root, then its conjugate $\bar{r} = \lambda - \mathrm{i}\mu$ is also a root. The complex-valued solution pair $y = e^{(\lambda + \mathrm{i}\mu)t}$ and $y = e^{(\lambda - \mathrm{i}\mu)t}$ can be re-expressed as the real-valued solution pair:

$$y = e^{\lambda t} \cos(\mu t) \quad \text{and} \quad y = e^{\lambda t} \sin(\mu t).$$

1.3 Non-Homogeneous Equations

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(t)$$

Then the general solution is given by

$$y(t) = y_c(t) + y_p(t),$$

where y_c is the complementary solution (homogeneous solution) and y_p is a particular solution.

1.4 Method of Undetermined Coefficients

1.5 Variation of Parameters

Theorem 1.6 (Variation of Parameters)**.**
Suppose $g, p_0, \dots, p_{n-1}$ are continuous on an interval I. Then a particular solution to the non-homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t) y^{(n-1)} + \dots + p_1(t) y' + p_0(t) y = g(t)$$

can be found by

$$y_p(t) = u_1(t) y_1(t) + u_2(t) y_2(t) + \dots + u_n(t) y_n(t)$$

where $y_1, y_2, \dots, y_n$ is a FSS to the corresponding homogeneous equation and $u_1, u_2, \dots, u_n$ are functions given by

$$u_i(t) = \int \frac{g(t) \cdot \det[M_i(t)]}{W(y_1, \dots, y_n)(t)} \, \mathrm{d}t \,, \qquad \text{for} \quad i = 1, 2, \dots, n, \quad \text{and}$$

$$M_i = \begin{pmatrix} y_1 & \cdots & y_{i-1} & 0 & y_{i+1} & \cdots & y_n \\ y_1' & \cdots & y_{i-1}' & 0 & y_{i+1}' & \cdots & y_n' \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & \cdots & y_{i-1}^{(n-1)} & 1 & y_{i+1}^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}.$$

Let $u_1, u_2, \dots, u_n$ be functions satisfying the system of equations

$$\begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \cdots & y_n^{(n-2)} \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \\ \vdots \\ u_{n-1}' \\ u_n' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ g(t) \end{pmatrix}$$

Then it can be verified that $y_p = u_1 y_1 + \cdots + u_n y_n$ is a particular solution to the non-homogeneous equation.

2 First order ODE systems

The general system of first order equations involving n dependent variables defined on an interval $I \subseteq \mathbb{R}$ can be written as

$$\begin{cases} y_1'(t) = F_1(t, y_1, \dots, y_n) \\ y_2'(t) = F_2(t, y_1, \dots, y_n) \\ \qquad \vdots \\ y_n'(t) = F_n(t, y_1, \dots, y_n) \end{cases} \qquad \text{for} \quad t \in I. \tag{1}$$

$$\text{with initial conditions} \quad y_1(t_0) = x_1, \quad y_2(t_0) = x_2, \quad \dots, \quad y_n(t_0) = x_n,$$

The solution to the system can be expressed in vector form as

$$\boldsymbol{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix}$$

Theorem 2.1 (Existence and Uniqueness)**.**

For the system of first order ODEs eq. (1), if the functions $F_1, F_2, \dots, F_n$ and the partial derivatives $\dfrac{\partial F_i}{\partial y_j}$ for all pairs $i, j = 1, 2, \dots, n$ are continuous on a region R defined by

$$R := [t_0 - a, t_0 + a] \times [x_1 - b, x_1 + b] \times \cdots \times [x_n - b, x_n + b] \subseteq \mathbb{R}^{n+1},$$

where constants $a, b > 0$, then there exists a unique solution $\boldsymbol{y}(t)$ for $t \in (t_0 - h, t_0 + h)$ to the IVP, where

$$h = \min\{a, b/M\} \quad \text{and} \quad M = \max_{\boldsymbol{v} \in R}\{|F_1(\boldsymbol{v})|, |F_2(\boldsymbol{v})|, \dots, |F_n(\boldsymbol{v})|\}.$$

Any n-th order ODE can be rewritten as a system of n first order ODEs. Given an n-th order ODE

$$y^{(n)} = F(t, y, y', \dots, y^{(n-1)}),$$

setting $z_1 = y, z_2 = y', \dots, z_n = y^{(n-1)}$, we obtain the system

$$\begin{cases} z_1' = z_2 \\ z_2' = z_3 \\ \qquad \vdots \\ z_{n-1}' = z_n \\ z_n' = F(t, z_1, z_2, \dots, z_n) \end{cases}$$

which is a system of n first order ODEs.

Definition 2.1 (First-Order Linear System)**.**

A system of first order ODEs is called **linear** if it can be expressed in the form

$$\boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) + \boldsymbol{g}(t) \qquad \text{for} \quad t \in I,$$

where

$$\boldsymbol{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix} \qquad \boldsymbol{g}(t) = \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix} \qquad \boldsymbol{P}(t) = \begin{pmatrix} p_{11}(t) & p_{12}(t) & \cdots & p_{1n}(t) \\ p_{21}(t) & p_{22}(t) & \cdots & p_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1}(t) & p_{n2}(t) & \cdots & p_{nn}(t) \end{pmatrix}$$

Definition 2.2 (Homogeneous First-Order Linear System)**.**

A first order linear system is called **homogeneous** if $\boldsymbol{g}(t) = \boldsymbol{0}$ for all $t \in I$, i.e., it can be expressed as

$$\boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) \qquad \text{for} \quad t \in I.$$